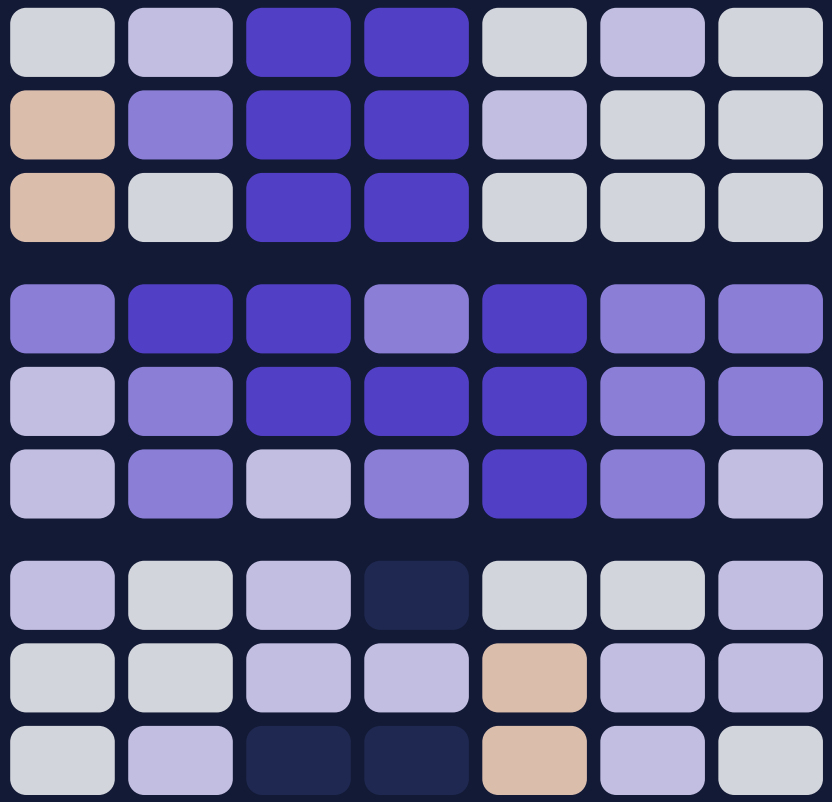
Texas MISA Fall 2026 Case Competition

Workforce Planner

Costco holiday-week staffing · Feb 12–18



Simulated data, not real Costco data

One holiday week, three scenarios

Presidents' Day window

Thu	Fri	Sat	Sun	Mon	Tue	Wed
Feb 12	Feb 13	Feb 14	Feb 15	Feb 16	Feb 17	Feb 18
	Valentine's Day			Presidents' Day		

Scenarios

BASE	Expected week
PEAK	Demand up to +30%, Feb 13–15
DISRUPT	Fewer staff, Warehouses 2 and 4

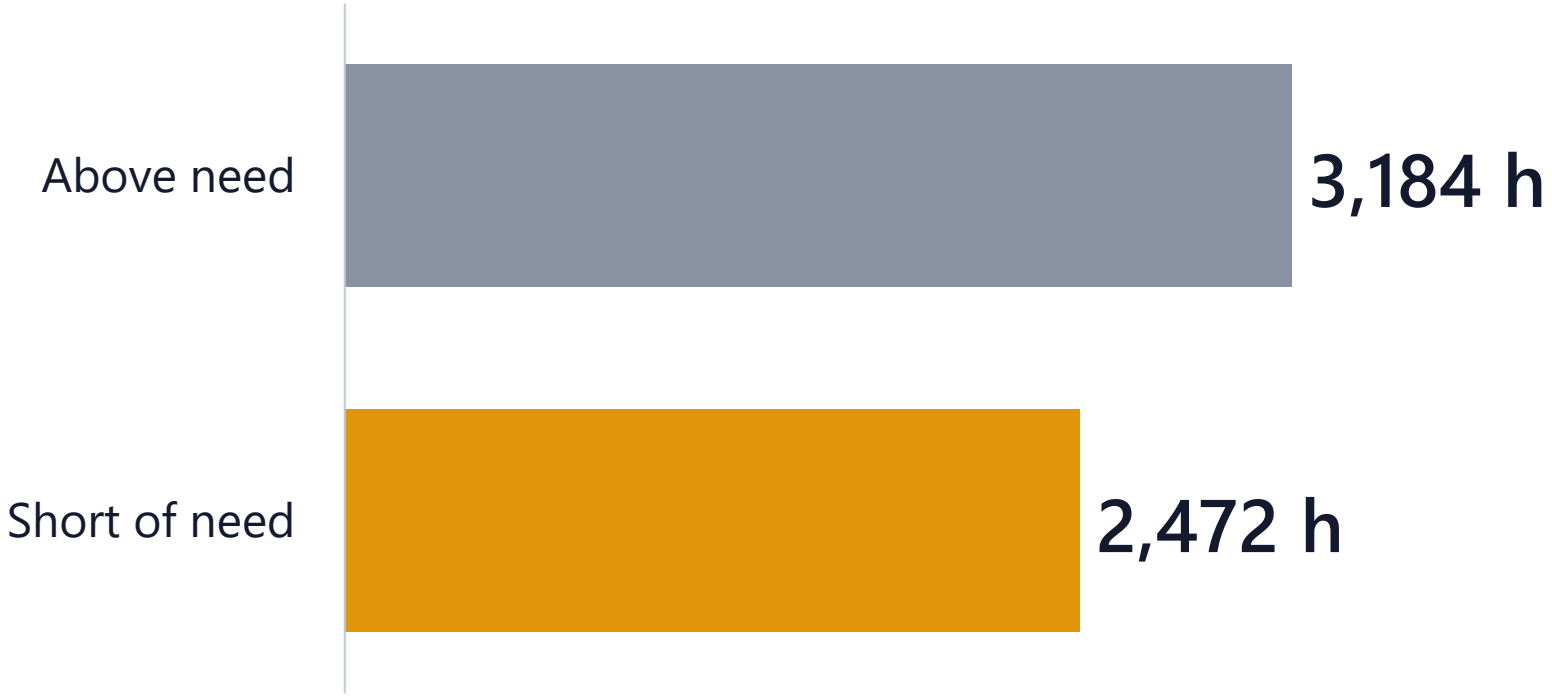
Service targets

Front End	**≤ 5 min**
Merchandising	**≤ 60 min**
Pharmacy	**≤ 20 min**

Simulated data, not real Costco data

Past schedules put hours in the wrong places

Hours against need, 6 weeks of history



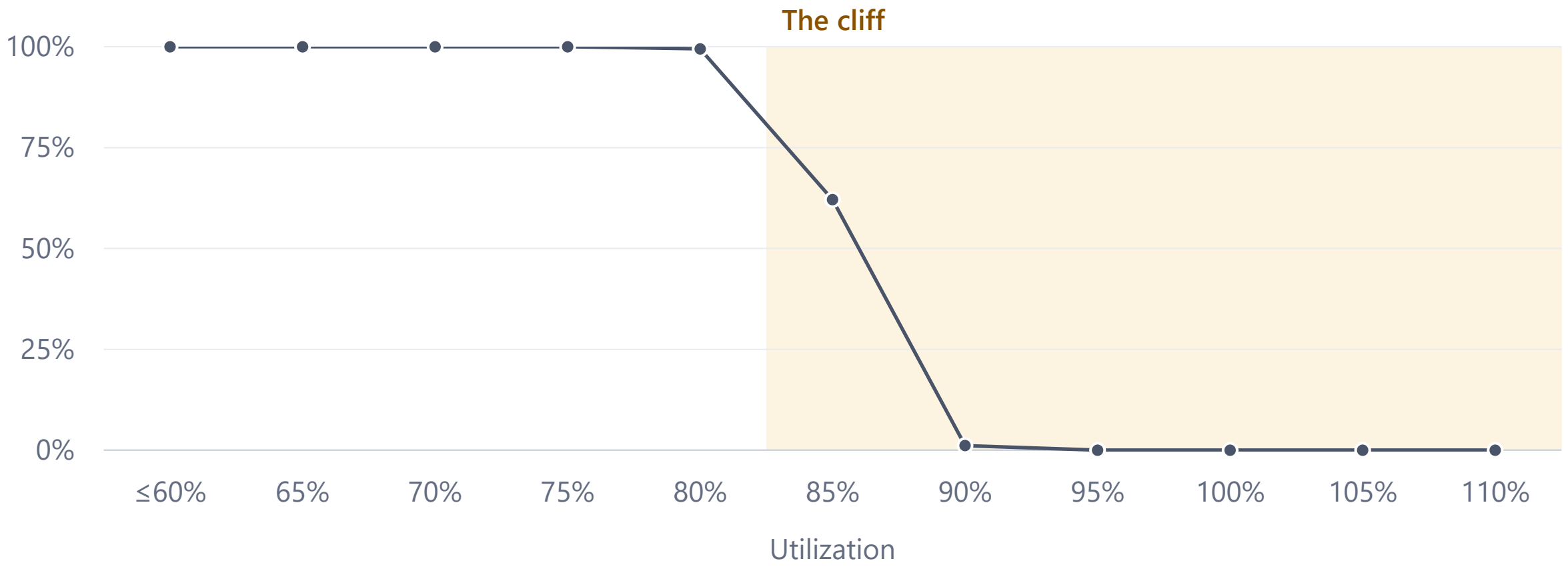
Net, all hours vs need

+1.2%

Simulated data, not real Costco data

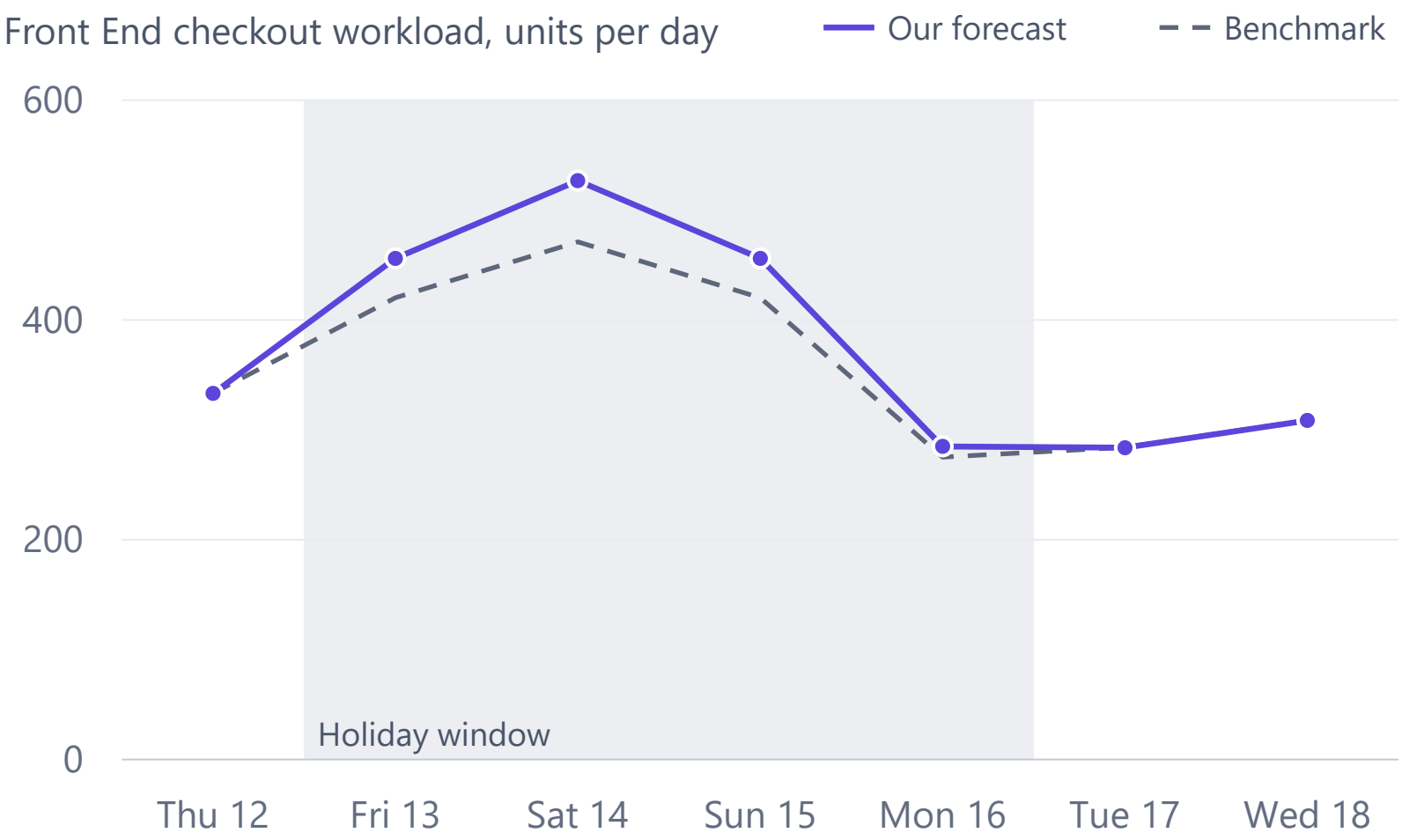
Service collapses past 85% utilization

Blocks meeting target, by utilization



Simulated data, not real Costco data

Demand peaks on the holiday weekend



Holiday-weekend lift

+12%

Daily error, held-out weeks

2.4%

vs 4.1% benchmark

Simulated data, not real Costco data

Five steps from forecast to plan



Forecast
Demand per block

Service risk
Chance of meeting target

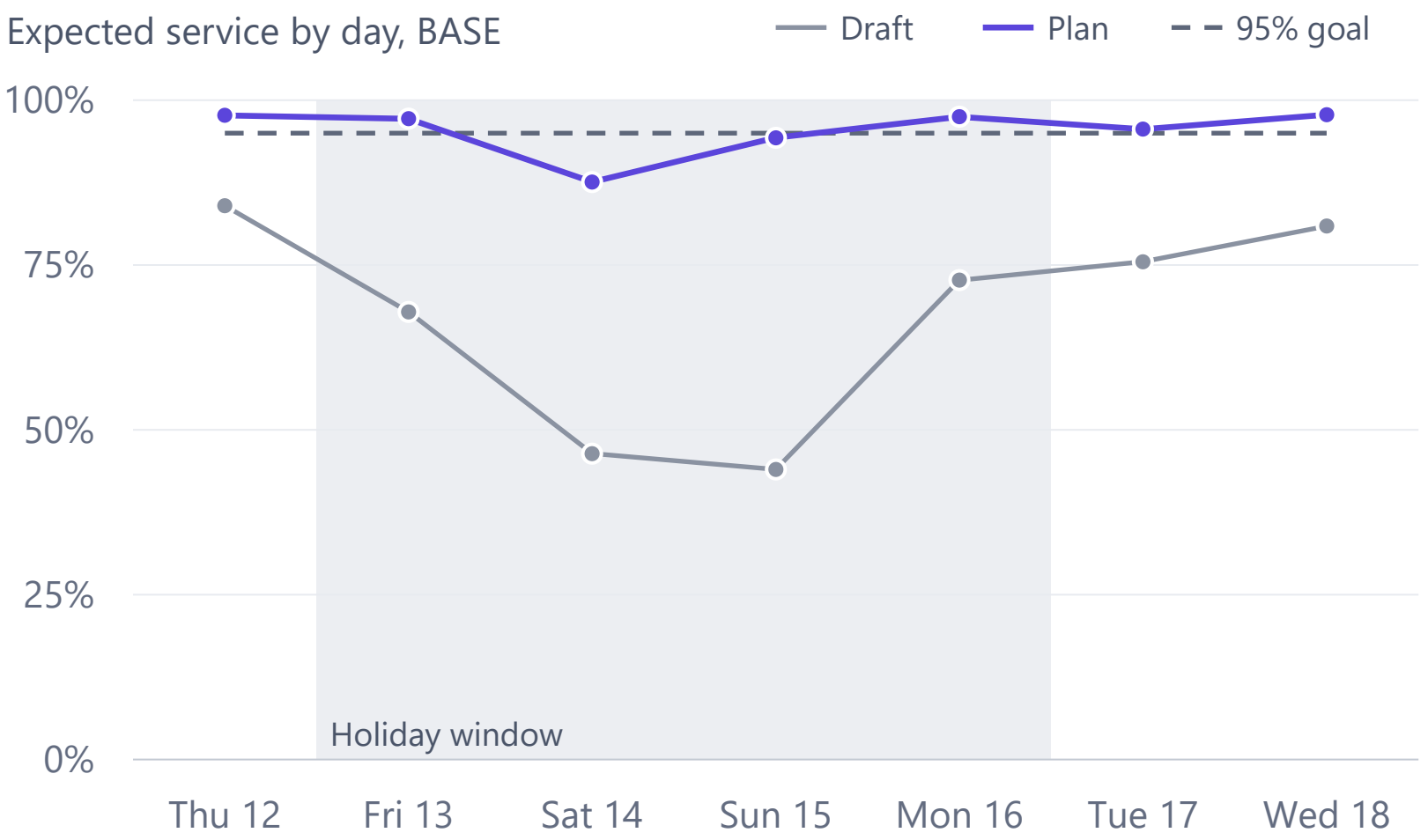
Cost rule
Regular, flex, overtime

Search
Every staffing pair

Explain
A reason per block

Simulated data, not real Costco data

Expected service rises from 68.5% to 95.6%



Labor cost

$377.5K

↗ +14.7% vs draft

Blocks below the goal

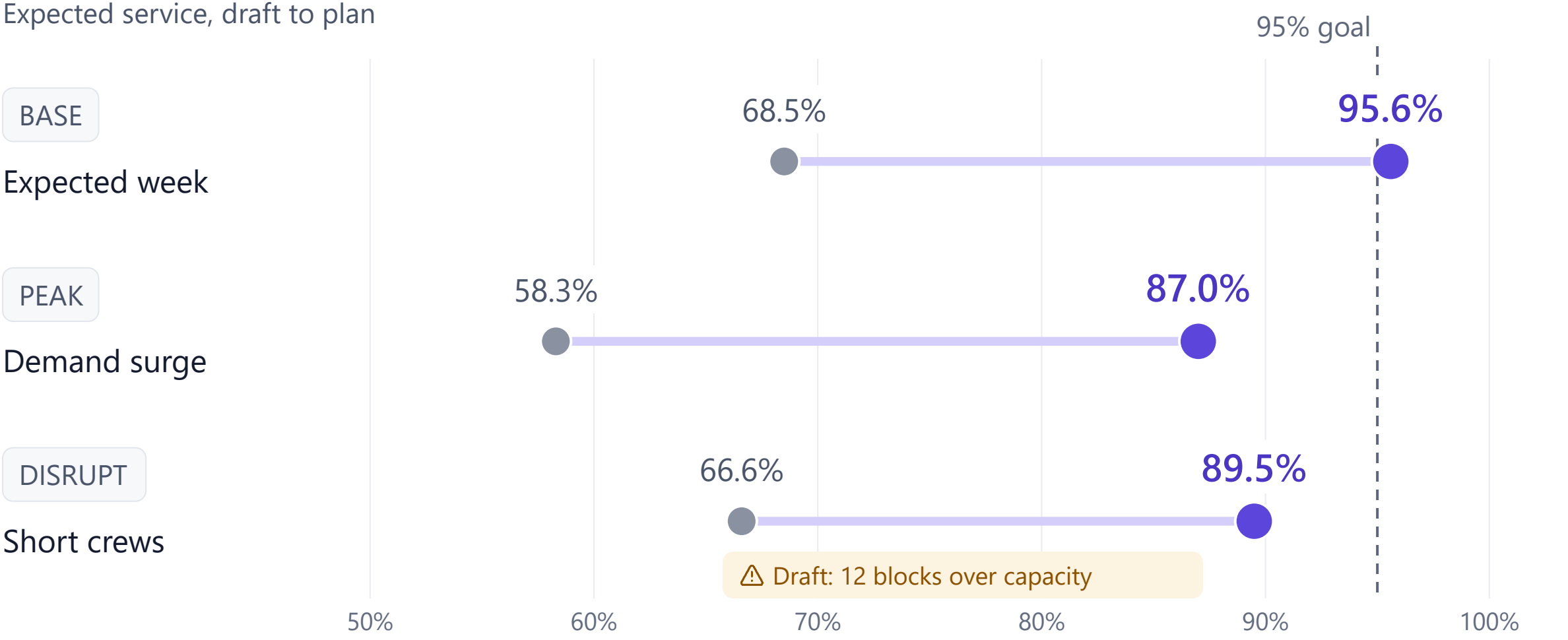
25

↘ −110 vs draft

Simulated data, not real Costco data

The plan lifts service in every scenario

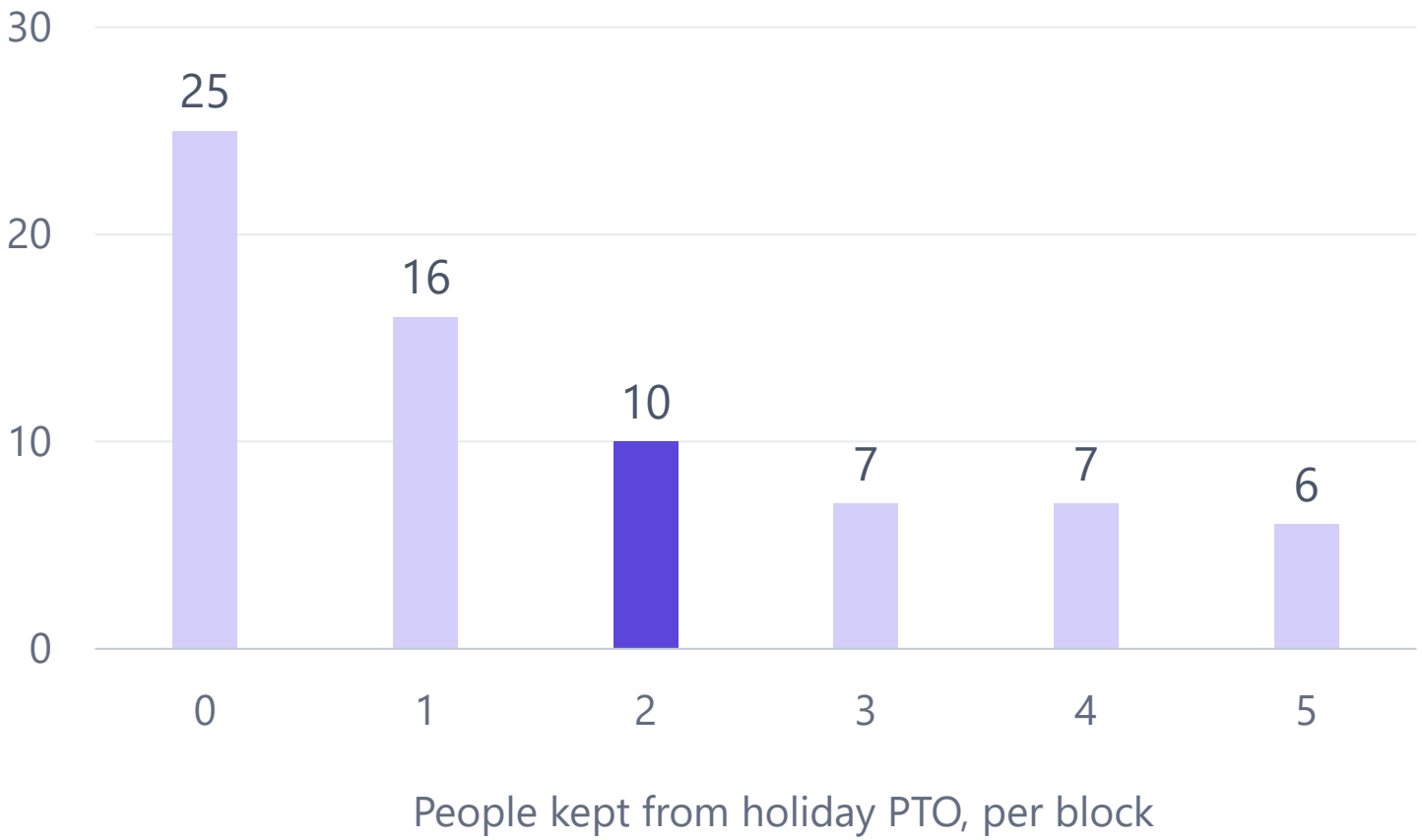
Expected service, draft to plan



Simulated data, not real Costco data

Limiting holiday PTO lifts service and cuts cost

Blocks below the goal, BASE



Keeping 2 per block, vs none

Blocks below the goal

−15

Expected service

+1.5 pts

Labor cost, week

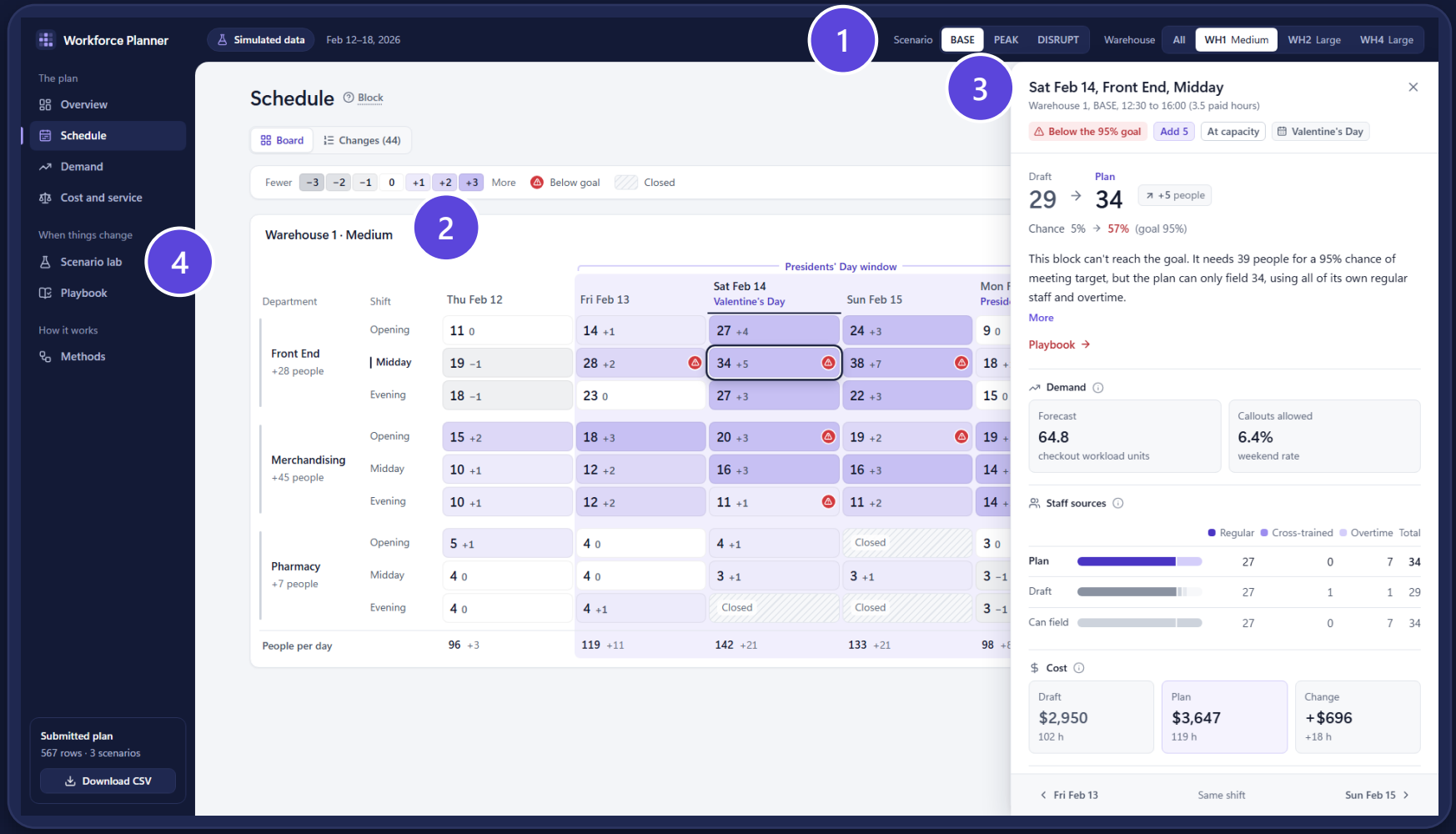
−$1.9K

Simulated data, not real Costco data

Live demo: the Workforce Planner

1. Scenario switch
2. The board
3. Block details
4. Scenario lab

▷ 5 minutes



Simulated data, not real Costco data

Adopt the plan and limit holiday PTO

+27.1 pts
Expected service, BASE

+14.7%
Labor cost, BASE

0
Blocks over capacity, any scenario

What we hand over

- **Staffing CSV**
- **Workforce Planner**
- **Report**
- **Notebook**

Simulated data, not real Costco data

Appendix

For questions

A1 Forecast method

A2 Forecast accuracy and its limits

A3 Service curve and risk model

A4 Allocation and cost rule

A5 Key assumptions, ours marked

A6 Cost and service against the draft

A7 Where capacity runs out

A8 Service goal and callout allowance

A9 Validation and checks

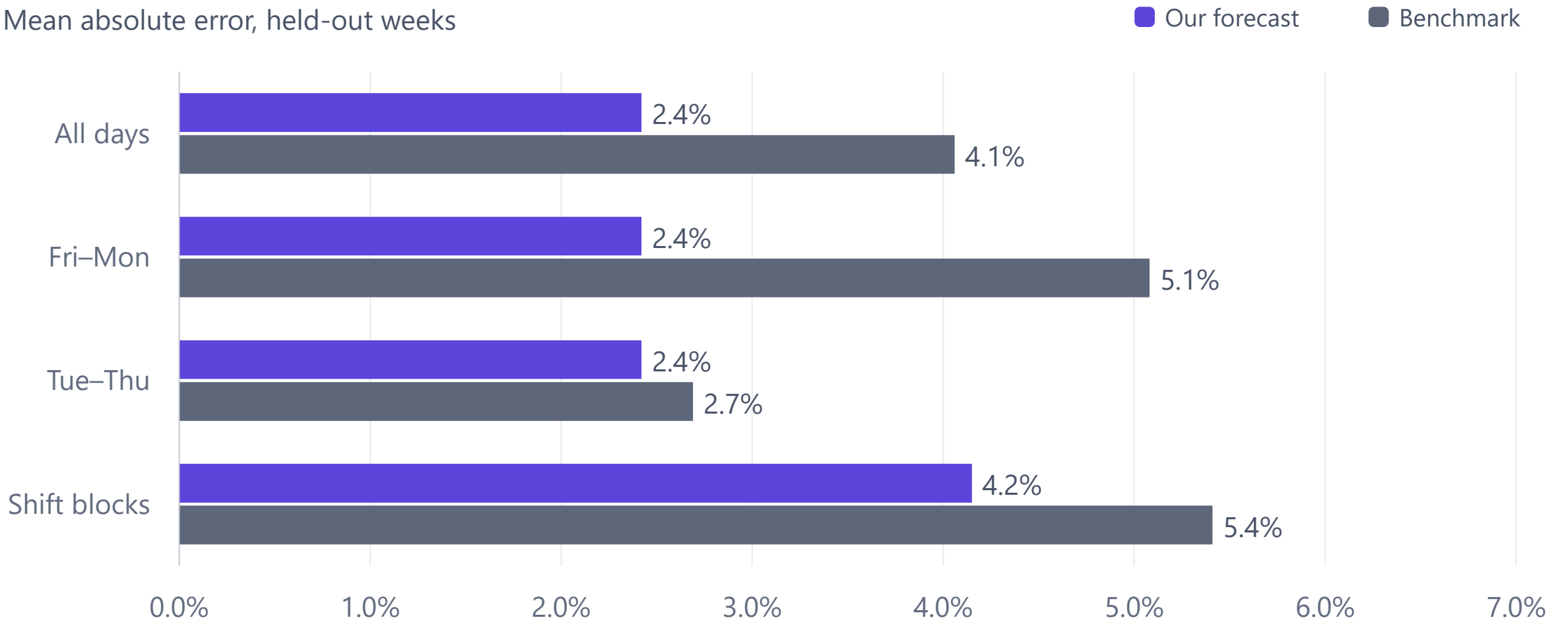
A10 Data notes

A11 The submission file

The forecast is a regression on the calendar

Driver	Evidence	In the plan
Weekday	In-sample R^2 0.977, with warehouse	In the model
Holidays	R^2 0.995 with holiday windows	Borrowed from MLK weekend
Weekly level	Held-out error 2.4%; levels 0.95 to 1.06	Last 4 weeks' average
Valentine's Day	No history to measure	+3%, Front End and Merchandising
Weather	r = −0.006 after weekday	Left out
Promotions	On every Fri–Sun and holiday (r = −0.017)	Calendar terms; PEAK's multipliers
Traffic	An outcome in the data pack	Left out

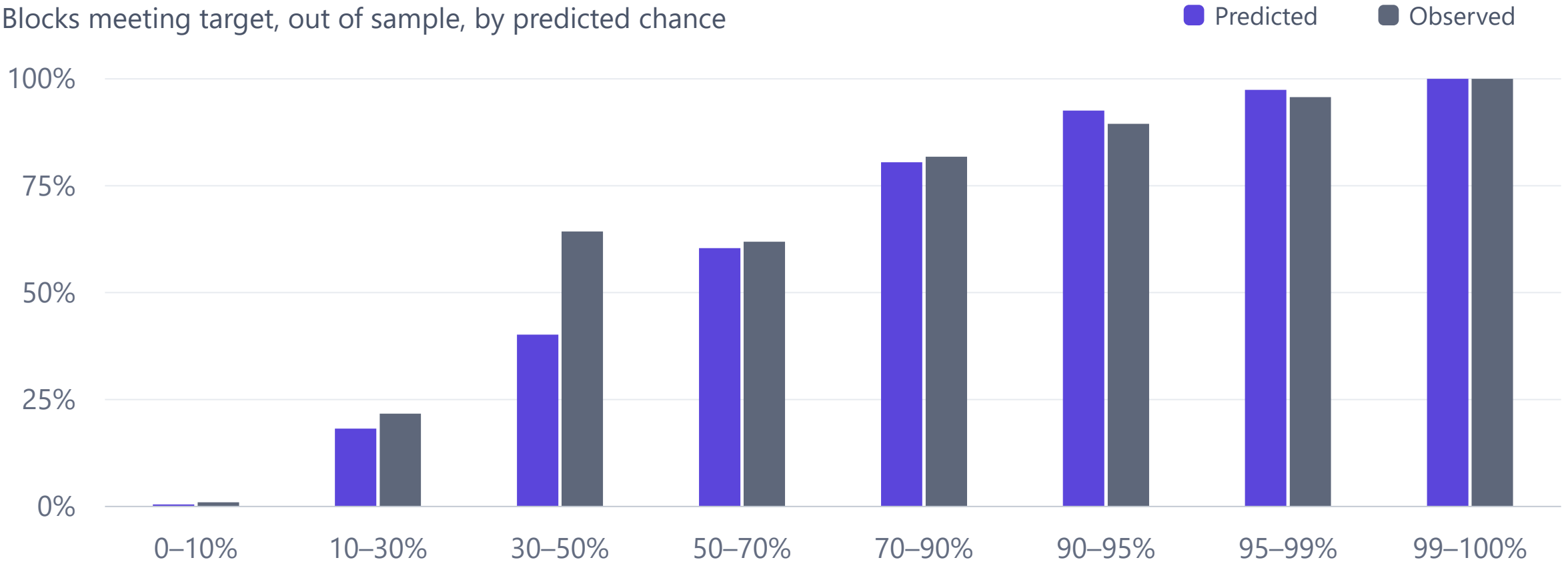
The forecast's edge comes from Friday to Monday



Simulated data, not real Costco data

Predicted chances match what happened

Blocks meeting target, out of sample, by predicted chance



Brier score **0.024** vs 0.177 for a flat rate · lower is better

Simulated data, not real Costco data

Cost follows the case's rule, solved per shift

Cost order

1 Regular staff
$28.75–$52.00 / h

2 Cross-trained
Partner's spare staff

3 Overtime
$37.95–$68.64 / h

Search order, Front End and Merchandising together

1 Highest chance
Up to the 95% goal

2 Fewest hours short
Of need

3 Lowest cost
In the case's cost order

BASE 975 h overtime · 95 h cross-trained

9 assumptions of our own, each with its basis

ID	Assumption	Basis	Whose
A-01	Cross-trained staff count only if the partner isn't using them	Our reading of the README	**Ours**
A-03	Plan for callouts: 5.2% call out, 49% backfilled	Six weeks of history	**Ours**
A-05	MLK weekend guides the Presidents' Day window	Same Friday-to-Monday shape	**Ours**
A-06	Valentine's Day +3%, Front End and Merchandising	No analog; tested	**Ours**
A-09	History's service curve holds in the planning week	The case says to estimate it	**Ours**
A-12	DISRUPT's illness brings flu-week callout rates	Seen together in January	**Ours**
A-13	Goal: a 95% chance in each block	Service is judged first	**Ours**
A-14	Next week's level: last 4 weeks' average	Best in the backtest	**Ours**
A-16	Some holiday-weekend PTO can still be limited	Approvals aren't in the data	**Ours**
	Labor standards, wages, shift hours, minimum staff	Data pack	Case
	Cost order: regular, cross-trained, overtime	Case, page 7; README	Case
	Scenario demand and availability	Planning grid	Case

Service rises in every scenario and department

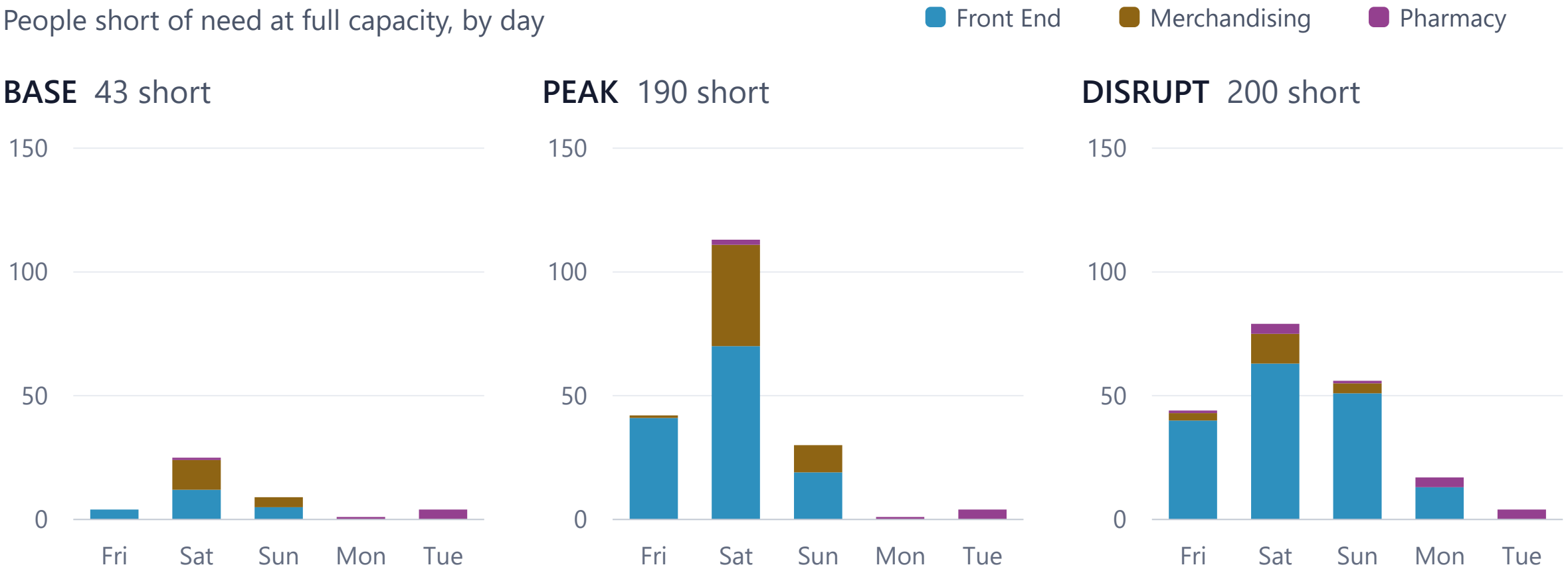
Scenario	Department	Hours draft	Hours plan	Cost draft	Cost plan	Cost change	Service draft	Service plan
BASE	**All**	**10,344**	**11,658**	**$329.1K**	**$377.5K**	**+14.7%**	**68.5%**	**95.6%**
	Front End	4,729	5,410	$139.2K	$163.3K	+17.3%	61.9%	**95.2%**
	Merchandising	4,846	5,404	$149.7K	$169.5K	+13.2%	58.5%	**95.0%**
	Pharmacy	770	844	$40.2K	$44.7K	+11.2%	87.7%	**96.6%**
PEAK	**All**	**10,344**	**11,923**	**$329.1K**	**$387.9K**	**+17.9%**	**58.3%**	**87.0%**
	Front End	4,729	5,494	$139.2K	$166.8K	+19.8%	49.1%	**79.4%**
	Merchandising	4,846	5,586	$149.7K	$176.4K	+17.8%	43.8%	**86.7%**
	Pharmacy	770	844	$40.2K	$44.7K	+11.2%	86.0%	**96.1%**
DISRUPT	**All**	**10,207**	**11,288**	**$326.4K**	**$366.6K**	**+12.3%**	**66.6%**	**89.5%**
	Front End	4,623	5,065	$137.0K	$152.7K	+11.4%	58.5%	**80.2%**
	Merchandising	4,834	5,392	$150.0K	$169.7K	+13.1%	58.3%	**94.5%**
	Pharmacy	750	832	$39.3K	$44.2K	+12.5%	85.6%	**94.5%**

DISRUPT draft cost: fieldable staff only; **36 people in 12 blocks can't be fielded**

Simulated data, not real Costco data

Capacity runs out on the holiday weekend, most on Feb 14

People short of need at full capacity, by day

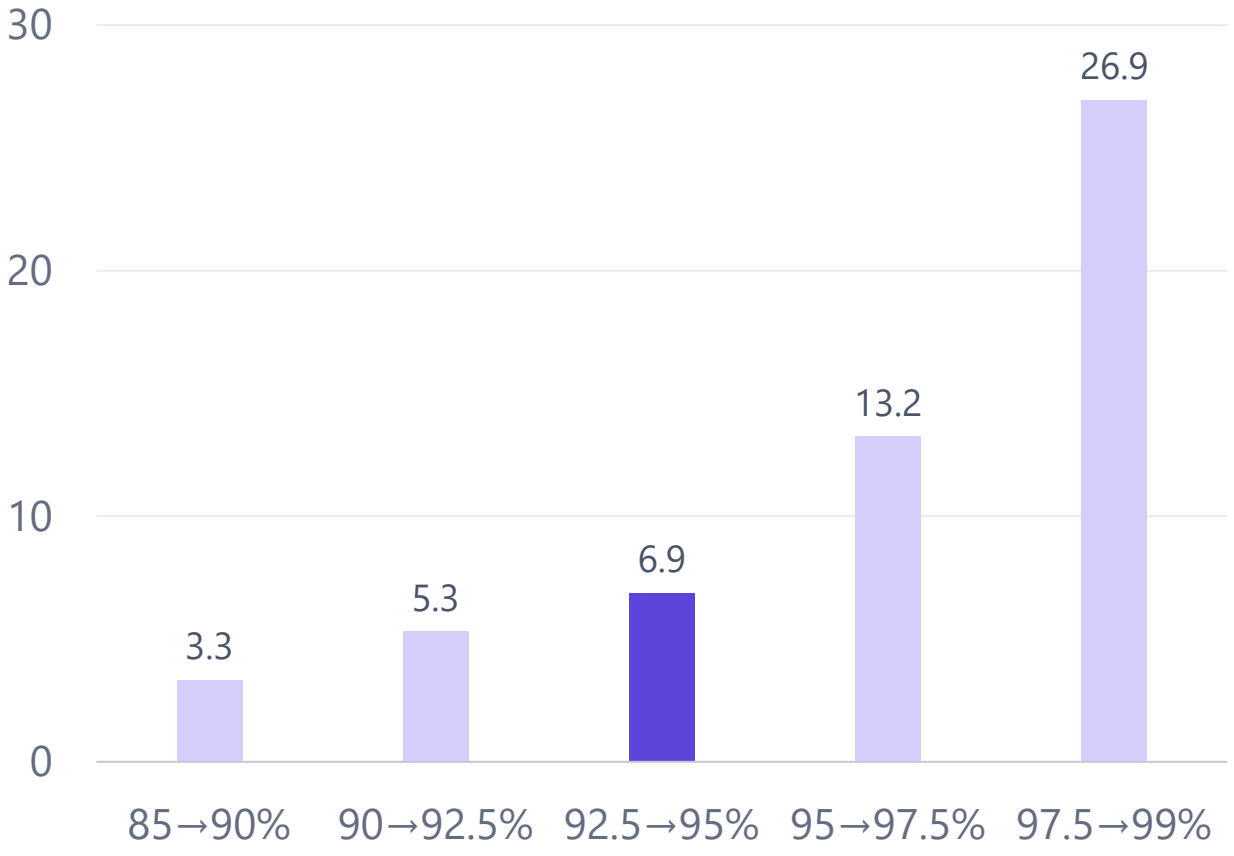


Feb 14, BASE: Front End and Merchandising at capacity **11 of 18 blocks** · Holiday PTO **4.2 fewer Front End staff per block**

Simulated data, not real Costco data

Each extra point of service costs more than the last

Labor cost per point of expected service, $K, BASE



Expected service if these risks occur, BASE

Plan allows for	Cost	Both risks	Forecast error	Neither
Both risks (ours)	$377.5K	96%	100%	100%
Forecast error	$354.8K	90%	98%	100%
Neither	$328.2K	73%	86%	99%
Current draft	$329.1K	69%	77%	80%

Simulated data, not real Costco data

Every check we could run passes

Submission file	567 rows on keys · 27 closed at 0 · 0 validator problems
Feasibility	All 540 open blocks within capacity and at or above minimum
Cost rule	Rebuilt row by row: every block and total to the half cent
Search	Independent brute force: same staffing in every shift checked
Risk math	Exact chance = Monte Carlo simulation
Calibration	Predicted 95–99%: met 95.7% (predicted 97.4%)
Explanations	Every quoted number tested, all 567 blocks
Reproducible	Pipeline rerun rewrites every output byte for byte

Simulated data, and the quirks that shaped the method

Simulated data	Not Costco data; mock wage rates
DISRUPT cuts more	36 of 189 blocks, cross-trained staff too: read from the grid
Promotions follow the calendar	Every Friday to Sunday and holiday: not separable
Traffic is an outcome	Never a model input
Holiday labels differ	MLK and Presidents' Day mapped to shared roles
One draft for all scenarios	The same 2,390 employee shifts in each

The submission file: all 567 blocks, joined on keys

A sample of rows, in the template's order

Scenario	Warehouse	Department	Date	Shift	Draft	Submitted
BASE	1	Front End	Feb 14	Evening	24	27
BASE	1	Front End	Feb 14	Midday	29	34
BASE	1	Front End	Feb 14	Opening	23	27
BASE	1	Pharmacy	Feb 14	Evening	0	0
BASE	1	Pharmacy	Feb 14	Midday	2	3
BASE	1	Pharmacy	Feb 14	Opening	3	4
DISRUPT	2	Front End	Feb 14	Evening	24	20
DISRUPT	2	Front End	Feb 14	Midday	31	29
DISRUPT	2	Front End	Feb 14	Opening	27	21

Rows **567 = 3 × 189** · Closed **27 at 0** · Value **total employees per block**